Human, Computer, and AI Interaction · Term Project 2026

Campus Services Helpdesk

Week 2 · Ideate with AI · Milestone 1 Draft

TRACK · Campus Services Helpdesk | HAI PRINCIPLE · Transparency

Main Idea

Our team proposes a bilingual Arabic-English campus helpdesk chatbot that answers student questions about registration, deadlines, and university rules using selected public university documents. It shows where each answer comes from and clearly says when the evidence is weak, conflicting, missing, or outside scope instead of guessing.

Project Concept & Approval Request

Problem. Students may have trouble locating current rules in university documents and checking whether an answer is trustworthy.

Proposed solution. A source-grounded chatbot that lets students inspect the document passage behind an answer. It will not access student records or make individual decisions.

Transparency in the interaction. Students can see the system's document scope, answer sources, and limits. When evidence does not support a reliable answer, the chatbot explains that and stops short of a guess.

Approval request

We request approval to develop this idea using a selected set of public university documents. We will confirm the source collection and any official contact route before implementation. Our first prototype will focus on registration, deadlines, and general university rules.

Source plan: the exact institution and public documents have not yet been selected; this draft makes no claims about real university policies.

Submission scope: Week 2 / Milestone 1 only. Later deliverables such as personas, metrics, architecture, and prototype are not included.

Two AI-Generated Concept Drafts

Concept Draft 1 · OpenCode

Evidence-First Helpdesk

Makes the retrieval process and evidence visible throughout the conversation.

1. Disclose chatbot scope and limits.
2. Search the indexed public documents and show the search scope.
3. Answer from retrieved passages with citations and a plain-language evidence state.
4. Show weak, missing, or conflicting evidence without guessing.
5. Decline out-of-scope questions; offer a human route only if configured.
6. Allow source inspection and optional problem feedback.

The full output also proposes language controls, feedback logging, a session summary, and a prototype outline.

Concept Draft 2 · Claude

“Show Your Source”

Builds the student journey around checking each answer against a visible source.

1. Introduce chatbot limits and document scope.
2. Ask in Arabic or English and show the search state.
3. Answer with an evidence card: source, section, date, link, and quote.
4. Let the student open and verify the original passage.
5. Explain weak evidence and out-of-scope refusals.
6. Offer correction or official contact when available.

The full output also proposes source reporting, bilingual source comparison, and saving cited answers.

Side-by-Side Comparison

Criterion	Evidence-First · OpenCode	“Show Your Source” · Claude
Strengths	Detailed weak/no-evidence and conflict paths; visible search scope; explicit no-guessing behavior.	Clear student journey; strong evidence panel and source inspection; correction/contact options.
Weaknesses	12 steps and extras may exceed first-prototype scope; evidence labels need a defensible rule.	Contacts, dates, and authoritative language versions require verified university information.
Feasibility	Feasible when reduced to retrieval, citations, and fallback.	Feasible when source cards use available metadata; defer bilingual comparison unless both versions exist.
Direction	Combine the shared core: source-grounded answers, visible citations, explicit uncertainty, and a clear out-of-scope response. Defer optional features.	

The concepts above summarize the two complete AI outputs preserved in the appendix.

Selected Core Interaction Flow

Step	User / system interaction
1 · Ask	Student chooses Arabic or English and asks about registration, a deadline, or a university rule.
2 · Retrieve	System searches only the public university documents selected by the team.
3 · Answer with evidence	Show a concise answer with the supporting passage and available source details.
4 · Weak or conflicting evidence	Explain what is missing or conflicting; do not fill gaps with general knowledge.
5 · Out of scope	Explain why the system cannot answer; do not infer personal records or decisions.
6 · Verify / refer	Let the student open the source. Show an official contact route only after verification.

For instructor feedback: confirm the document collection and whether an official-contact card belongs in the prototype.

Week 2 Check-In

Present the idea, both concept drafts, the strengths/weaknesses comparison, and the selected direction. Be ready to explain the two tools, prompts, and team revisions.

Reference: HAI Term Project 2026 file, Week 2 “Ideate with AI,” p. 7; Appendix A: AI Prompt Log, p. 13. Week 2 asks for two concepts from different AI tools, a short comparison, prompt log entries, and the idea for approval.

AI Prompt Log

Prompts are reproduced from the two texts supplied earlier for this project. The full tool outputs are preserved in the following appendix pages.

Entry 1 · OpenCode · Exact prompt

You are a human-AI interaction designer. Draft a core interaction flow for a bilingual Arabic-English Campus Services Helpdesk chatbot for university students. The chatbot answers questions about registration, deadlines, and university rules using only retrieved passages from public university documents. The HAI principle is Transparency. For each step, show what the user does, what the system does, what evidence or source is shown, and what happens when evidence is weak or the question is out of scope. Keep the flow feasible for a student team to prototype. Do not invent university policies, source documents, or tool outputs. Format as numbered steps.

What it gave	A 12-step evidence-first flow covering scope, bilingual controls, visible retrieval, citations, conflict/weak/no-evidence paths, out-of-scope handling, escalation, source inspection, feedback, and session summary.
Team changes and why	Keep scope disclosure, retrieval, citations, weak/conflicting evidence, out-of-scope handling, and source inspection in the first prototype. Defer feedback logs, session exports, advanced highlighting, and complex confidence rules until feasibility and the source data are verified.

Entry 2 · Claude · Exact prompt

Propose a second, distinct concept for the same project: a bilingual Arabic-English Campus Services Helpdesk for student questions about registration, deadlines, and university rules, grounded only in public university documents and aligned with Transparency. Focus on making the system’s limits and uncertainty clear. Describe the user journey from asking a question to verifying the answer, including source display, weak-evidence handling, out-of-scope handling, and a way to reach the relevant campus office. Do not invent real university facts. Format as a compact storyboard with user and chatbot actions.

What it gave	A nine-step “Show Your Source” storyboard covering scope, bilingual questions, visible search, evidence cards, verification, reporting, uncertainty, out-of-scope handling, and official contact.
Team changes and why	Keep the evidence card and source verification. Show dates, authoritative-language claims, or office contacts only when confirmed in selected documents. Defer correction logs and saved-session features from the minimum prototype.

Record integrity: if the prompts entered in the tools differ from these texts, replace them with the exact versions from the tool histories.

Prompt-log fields follow Appendix A: tool, exact prompt, what it gave, and what the team changed and why.

Output 1 · OpenCode

Full output as supplied in the attached scope note. Bracketed values are placeholders, as stated in the source.

Scope note: all brackets ([…]) are placeholders for what the retrieval pipeline actually returns. No policy text, document titles, dates, or tool outputs are assumed.

1. Session start — scope & limits banner

User: Opens the chatbot; picks language (العربية / English), or it defaults from browser locale.

System: Renders UI, labels, and fonts fully in the chosen language; shows a persistent one-line scope banner: answers come only from retrieved passages in public university documents; no advice beyond them.

Evidence shown: Banner + a "How this works" link explaining retrieval, citation, and confidence labels (the mechanism, not a policy claim).

Weak / out of scope: If the student dismisses the banner, the scope line remains visible as a collapsed chip so limits are never fully hidden.

2. Language control stays live

User: Taps the language toggle at any turn.

System: Switches the interface and re-renders the current answer and citations in the other language; retrieved passage text stays in its original language with a label marking it as untranslated source text.

Evidence shown: Original passage kept verbatim beside the translated rendering, tagged "original" / "translation".

Weak / out of scope: If machine translation of a passage is uncertain, show the original plus a note that the translation may be imperfect — never silently paraphrase.

3. Question intake

User: Asks a question about registration, deadlines, or university rules in either language.

System: Classifies the query as in-scope or out-of-scope before answering; for out-of-scope, it does not attempt an answer (step 8).

Evidence shown: Nothing yet — the student sees a "searching public university documents…" state, so no answer appears without visible retrieval behind it.

Weak / out of scope: Unrecognized query → routed to step 8 instead of guessing.

4. Retrieval with visible search scope

System: Searches only the indexed corpus of public documents; shows what was searched (e.g., "searching N public documents: list of titles"). No external web search, no hidden knowledge.

Evidence shown: Corpus list and result count per query, expandable.

Weak / out of scope: Zero hits → skip to step 7 (no evidence). Corpus list itself is shown so the student can judge coverage.

5. Answer composed from passages only

System: Generates the answer using only retrieved chunks; prompt constrains it to cite [1], [2] inline for every factual

claim; claims with no citation are dropped by a post-check.

Evidence shown: Answer block + evidence panel with, per citation: document title, section/page, direct quote of the retrieved passage, link to the public document, retrieval date, and a strength label (Strong / Partial / Weak / None) derived from simple, stated rules (number of supporting passages, whether passages agree, document recency).

Weak / out of scope: Partial or Weak → answer is rendered under a visible amber banner: "Evidence is incomplete — read the passage before relying on this." The system must not fill the gap with general knowledge.

6. Conflicting or ambiguous evidence

System: Detects disagreement between passages (different dates, different wording).

Evidence shown: Both passages side by side with their document titles and dates; label "Sources conflict"; no automatic resolution, no "best guess."

Weak / out of scope: Student is offered two routes: view both sources, or escalate (step 9).

7. Weak-evidence / no-evidence path

User: Receives an explicit non-answer.

System: States plainly: "I could not find a passage in the public documents that answers this." Optionally offers the closest partial passage(s) labeled as related, not an answer.

Evidence shown: Empty-state with the search scope from step 4, plus the near-miss passages and why they were not treated as answers.

Weak / out of scope: No improvisation. The turn ends with the escalation options from step 9.

8. Out-of-scope path

System: Names the category of what was asked (e.g., personal records, individual case decisions, anything not covered by the indexed documents) and explains why it can't answer — without inventing where the answer can be found.

Evidence shown: Scope banner restated; link back to "How this works."

Weak / out of scope: Never fabricates a plausible answer; offers escalation instead.

9. Escalation to a human / official channel

System: Shows a contact or referral option only if such a channel exists in the prototype's config; otherwise shows: "No human handoff is configured in this prototype."

Evidence shown: Copy of the student's question + the citations (or the absence-of-evidence record) packaged as a handoff summary, labeled "Draft — verify before sending."

Weak / out of scope: Works identically for weak-evidence, conflict, and out-of-scope turns.

10. Source inspection (on-demand)

User: Taps any citation or "Show sources."

System: Opens the full retrieved passage plus document metadata, and the public URL; a diff-style highlight marks the exact lines used for the answer.

Evidence shown: Passage, quote highlight, title, section, date, link.

Weak / out of scope: If the passage no longer resolves or the link is broken, show that explicitly rather than hiding the

citation.

11. Feedback & correction loop

User: Rates the answer (helpful / not helpful) and can flag "wrong citation," "sources conflict," or "not covered by documents."

System: Logs the flag with the session's citations for a human to review; does not auto-retrain or auto-edit anything.

Evidence shown: Confirmation message stating exactly what was recorded and that a human will review it.

Weak / out of scope: Flags on Weak/None turns are tagged separately so coverage gaps are visible to the team.

12. End of session — transparency summary

System: Produces a recap: questions asked, answers given, strength label per answer, full citation list, and a plain statement of what remains unverified.

Evidence shown: Exportable summary (copy/download) containing only the citations and labels.

Weak / out of scope: Any turn with Weak/None evidence is listed under "Needs verification."

Prototype feasibility (student team): static corpus of the university's public documents converted to text → keyword/BM25 or a hosted embedding index → prompt-constrained generation with inline citation requirement → post-check that drops uncited claims → simple strength rules (passage count, agreement, date) → Streamlit or plain web UI with a single language toggle. Steps 1, 4, 5, 7, and 8 are the minimum viable transparency core; 6, 9, 10, and 11 can be stubbed behind a config flag.

Output 2 · Claude

Full “Show Your Source” storyboard as supplied in the conversation.

Concept 2: "Show Your Source" Campus Helpdesk

*(I don't have your first concept in this conversation, so I've made this one distinct by its core idea: **every answer is a claim that must be checked against a visible public document, and the bot is designed to say "I can't confirm that" often and clearly.**)*

All facts below are placeholders in [brackets]. The real system would only use the university's public documents.

Storyboard

#	User action	Chatbot action	Transparency feature
1. Arrive	Opens the helpdesk and picks العربية or English (switchable anytime).	Shows a short "What I am / am not" card: *"I only read [University]'s public documents. I can't see your records, make decisions, or replace an office. My answers can be wrong or outdated."* Lists the document library with dates.	Limits stated **before** the first question
2. Ask	Types: *"What is the last day to drop a course?"* (or the Arabic equivalent, or a mix of both).	Replies in the user's language. Shows status: *"Searching [N] public documents…"*	Visible retrieval step, not a black box
3. Answer with strong evidence	Reads the response.	Gives a short answer, then an **Evidence box**: document title, section/page, publication date, link, and the exact quoted passage. Adds a label: **"Direct match: the document states this explicitly."** Ends with *"Check the date: [Document, version date]."*	Source display; evidence level in plain words
4. Verify	Taps "View in original."	Opens the source passage highlighted, in its original language. If the document exists in both languages, offers a side-by-side view and notes which version the university lists as authoritative: *"If the versions differ, [the authoritative version] applies."*	Translation risk disclosed
5. Report a problem	Taps "This looks wrong or outdated."	Logs the question, answer, and source; confirms receipt; shows the public corrections log. Offers the office contact (frame 8).	Accountability loop
6. Weak evidence	Asks: *"Can I retake a course more than twice?"*	The bot finds only an indirect or partial passage. Label: **"Partial evidence: I could not find a rule that directly answers this."** It shows what it did find, labeled *"Related, not conclusive,"* and states what is missing. For conflicting	Uncertainty shown as a state, not hidden in tone

		or old sources it shows both side by side with dates and says *"These sources disagree. I can't tell which applies."*	
7. Out of scope	Asks: *"Will my appeal be approved?"* or *"What's my GPA?"* or *"Is [other university] easier?"*	Declines in one sentence with a reason: *"I can't see personal records," "I can't predict decisions,"* or *"My documents only cover [University]."* Then offers what it can do: *"I can show the public appeal procedure."*	Refusal is explained, never silent or vague
8. Reach the office	Taps "Contact the right office."	Suggests the office based on the topic (e.g., [Registrar's Office]) as a card: name, email/phone/location, [office hours], and the document that names this office as responsible. Offers a **pre-filled draft** containing the question, the bot's answer, and the sources, with no personal data added automatically. Shows an "Escalate" option on every weak-evidence or out-of-scope screen.	Human route always available
9. Close	Ends the chat or rates it.	Shows a reminder: *"Confirm important decisions with the office. This chat is not an official statement."* Offers to save the answer with its sources and date.	Closing reminder; saved answer keeps its citation

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Evidence states (always shown with a label and icon, not color alone)

1. Direct match: a document explicitly states it.

2. Partial evidence: related text only; gaps listed.

3. Conflicting or outdated sources: both shown with dates.

4. No evidence found: *"I found nothing in the public documents,"* plus the office contact.

5. Out of scope: reason given, alternative offered.

Transparency principles in this design

• **Claim and evidence are inseparable:** no answer appears without a source box.

• **No guessing to fill gaps:** if the documents are silent, the bot says so.

• **Dates everywhere:** each source shows its version date, and the library shows when it was last updated.

• **Visible limits:** the scope card, evidence labels, and closing reminder repeat the same message at different moments.

• **Privacy by default:** no personal records accessed; the office draft contains only what the user chooses to send.

Open questions for the real project

• Which documents count as authoritative, and who maintains them?

• How often is the library refreshed, and how is that shown to users?

• Who reviews the corrections log, and how fast?